Has the performance gap between Formula 1 cars closed? A lap-time analysis of 1996–2024

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1 Introduction

Formula 1 is a sport in which the machine matters at least as much as the driver. Ror most of its history the grid has been sharply divided between a few teams and a tail of entries several seconds per lap slower. Since the 1990s the governing body has introduced restrictive technical regulations, standardised components and, from 2021, a budget cap, all with the stated aim of closing that gap. In this report i ask whether the gap has in fact closed, and if so, *why*: have the cars genuinely converged, or has the slow tail of the grid simply stopped entering? The two explanations predict different things and can be separated in the data. The analysis covers 1996–2024, the period for which individual lap times exist.

2 Data sample

I use the *Formula 1 World Championship* dataset, a public release of the Ergast motor-racing database hosted on Kaggle [1]. Four files are used: `lap_times.csv` (every individual lap, in milliseconds), `results.csv` (which driver drove which car), `races.csv` and `constructors.csv`. The championship goes back to 1950, but lap-by-lap timing begins only in 1996, which sets the start of the sample.

I applied three cleaning steps: lap times outside 30 s to 300 s were discarded as timing errors or cars limping to the pits; opening laps were removed, as they start from a standstill; and a constructor was counted in a race only if at least 20 timed laps were recorded for it, so that a car which crashed early cannot contribute a meaningless "best lap". After cleaning, **542 races** entered the analysis.

3 Analysis method

For each race and each constructor, I take the single fastest lap set by either of its cars. The minimum is used rather than the mean because a car's best lap is close to its true potential, whereas its average lap is contaminated by traffic, fuel load and pit stops. Two quantities are then formed per race:

$$G_{\text{full}} = 100\,\frac{t_{\text{slowest}} - t_{\text{fastest}}}{t_{\text{fastest}}}, \qquad G_{\text{trim}} = 100\,\frac{t_{\text{2nd slowest}} - t_{\text{fastest}}}{t_{\text{fastest}}} \quad [\%]. \tag{1}$$

G_{full} measures how wide the grid is; G_{trim} ignores the single slowest constructor and so measures how close the rest of the field is. If the two behave alike, the cars have converged; if only G_{full} falls, the change is in who enters, not in the machinery. Percentages are used so that circuits of different lap lengths can be compared.

Races are aggregated per season using the median, which is insensitive to the few wet or safety-car-dominated events that would distort a mean; the uncertainty on the median is $1.253\,\sigma/\sqrt{N}$. Season medians are then averaged within eras by inverse-variance weighting. The analysis uses `pandas` and `numpy`.

4 Results and discussion

Figure 1 shows both quantities per season. The field spread is clearly *not* a smooth function of time: a straight-line fit over the whole period returns a slope of -0.024 ± 0.026 percentage points per year, only 0.9σ from zero, with $\chi^2/\text{ndf} = 14.4$. A linear trend is simply the wrong model.

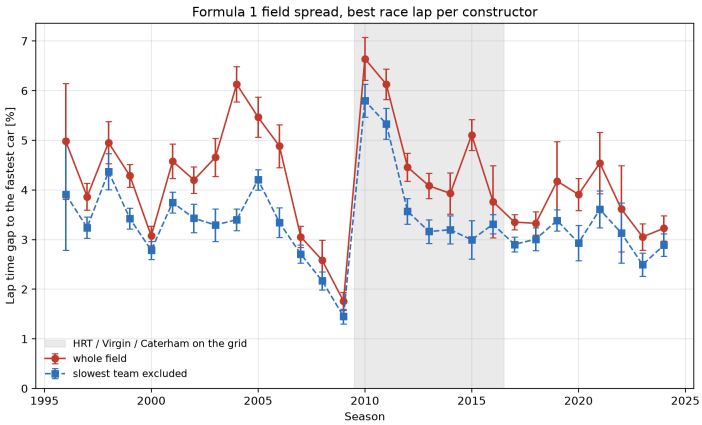


Figure 1: Season-median lap-time gap to the fastest car, for the whole field (red) and with the single slowest constructor excluded (blue). The shaded band marks the seasons in which HRT, Virgin/Marussia and Lotus/Caterham were on the grid.

The reason is visible in the figure. The gap peaks at $4.88 \pm 0.13\,\%$ during 2010–2016, the years in which three new teams entered the championship far off the pace, and this excursion cancels the decline on either side of it. Within eras the gap does fall: -0.154 ± 0.071 percentage points per year over 1996–2009 and -0.098 ± 0.037 over 2012–2024.

Comparing like with like, by setting the new-team years aside, gives the headline result (Table 1). For the whole field the gap falls from $4.00 \pm 0.08\,\%$ to $3.39 \pm 0.10\,\%$, a reduction of $0.61 \pm 0.13\,\%$, or 4.7σ. Excluding the single slowest car, however, the same comparison gives only $0.22 \pm 0.11\,\%$ (2.1σ). The bulk of the narrowing is therefore carried by the slowest constructor alone: the tail of the grid has disappeared, while the remaining cars have converged only marginally. The two differences are correlated, being measured on the same races, so their gap indicates size rather than an independent significance.

	1996–2008	2010–2016	2017–2024
Whole field [%]	4.00 ± 0.08	4.88 ± 0.13	3.39 ± 0.10
Slowest team excluded [%]	3.20 ± 0.06	3.74 ± 0.10	2.98 ± 0.08

Table 1: Inverse-variance weighted mean of the season medians in each era.

Two caveats remain. The metric uses race laps, where a slow car in clear air can out-pace a fast car in traffic; qualifying times would be cleaner but do not cover the full period. And the number of constructors varies between 10 and 13, which matters because a maximum over more teams is larger on average – part of why G_{trim} is the more stable quantity.

Conclusion. The gap between the fastest and slowest Formula 1 car has not shrunk steadily since 1996; it is governed by whether backmarker teams are present, and widened to $4.88 \pm 0.13\,\%$ when three of them joined in 2010. Comparing grids of similar composition, the field is $0.61 \pm 0.13\,\%$ tighter in 2017–2024 than in 1996–2008 (4.7σ), but only $0.22 \pm 0.11\,\%$ of that survives when the slowest car is removed. Most of the convergence is the disappearance of the slow tail rather than the cars themselves growing alike.

References

[1] V. Rao, *Formula 1 World Championship (1950–2024)*, Kaggle, derived from the Ergast Developer API, `https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020` (accessed October 15, 2026).